National University of Computer and Emerging Sciences

Electro Mechanical Systems (EE2015/2010)

Date: April 07th , 2026

Course Instructor(s)

Mr. Muhammad Abdul Majid

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Sessional-2 Exam

Total Time: 1 Hours

Total Marks: 50

Total Questions: 02

Semester: SP-2026

Campus: Lahore

Deptt: Electrical Engineering

M.Waleed Naeem 24L6023 4B-1

Student Name Roll No Section Student Signature

Vetted by Vetter Signature

Instructions:

Please **do not** ask any question, in case of an ambiguity write down your valid assumption and solve accordingly.

Show all the working and box out important answers.

National University of Computer and Emerging Sciences

CLO #3:

Apply operating principles to model and solve performance- related problems of DC machines

Q1: **[marks =25]**

A 22HP, 250V compensated DC shunt motor has armature resistance of 0.3 Ω. Its field resistance is 100 Ω and adjustable resistance connected in series with field coil can vary from 55 Ω to 257 Ω. Its no load characteristic curve at a speed of 1000 rpm is tabulated as under.

I_F (A)	0.3	0.6	0.9	1.2	1.4	1.5
E_A (V)	135	235	274	292	295	297

Apply the operating principles and model of DC machine to find out following quantities assuming rated terminal voltage in all cases.

a) No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 127 ohm.
b) When the motor is loaded it absorbs a line current of 61.2A, assuming same field current as in part (a), determine the speed at this loaded condition
c) Power converted from electrical to mechanical in part b.
d) Induced torque in part b
e) Maximum possible no load speed

CLO #4:

Analyze the construction, operation, and performance characteristics of synchronous machines under various loading and excitation conditions

Q2: **[marks 25]**

A 2500-V, 1200-kVA, 0.8-PF-lagging, 60-Hz, 4 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2 Ω and an armature resistance of 0.2 Ω. At 60 Hz, its friction and windage losses are 30 kW, and its core losses are 20 kW. The field circuit has a dc voltage of 220 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 22 to 200 Ω.

The generator is providing 713.75 KVA at (line to line) load voltage of 2235.2V and power factor of 0.866 lagging. OCC of the generator is given below.

Analyze the operation of synchronous generator and find out the following quantities.

a) Internally generated voltage
b) Voltage regulation
c) Value of field current
d) Efficiency
e) Output torque of prime mover

I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2600	2750	2850